

Section: Particles and Waves

Topic: Forces on Charged Particles

An electron travels through a uniform electric field between two parallel plates.

- Plate separation: $d = 0.015 \text{ m}$
- Voltage across plates: $V = 4.5 \text{ V}$
- Electron charge: $e = 1.6 \times 10^{-19} \text{ C}$

(a)

Calculate the magnitude of the electric field between the plates.

Use:

$$E = \frac{V}{d} = \frac{4.5}{0.015} = 300 \text{ V/m}$$

 *Revision tip:*

Electric field strength is voltage per metre:

$$E = \frac{V}{d}$$

(b)

Calculate the magnitude of the force on the electron in this electric field.

Use:

$$F = Eq = 300 \times 1.6 \times 10^{-19} = 4.8 \times 10^{-17} \text{ N}$$

 *Revision tip:*

The force on a charged particle in an electric field is given by:

$$F = Eq \text{ — like } F = ma, \text{ but with charge instead of mass.}$$